

LYDENBURG, South Africa

WMO: 681850

Lat: 25.1119S

Lon: 30.4767E

Elev: 4701

StdP: 12.36

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	36.7	38.9	15.9	14.6	56.6	19.8	17.7	55.5	N/A	N/A	N/A	N/A	N/A	N/A	0.325

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.8	84.7	62.1	82.6	62.1	80.7	62.4	68.6	77.4	67.5	76.2	66.5	75.1	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.2	115.1	72.5	64.8	109.7	71.4	63.8	105.6	70.7	36.0	77.3	35.0	76.2	34.1	75.0	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.3	69.7	69.5	67.2	63.4	58.9	55.1	53.8	58.7	63.5	65.0	66.4	68.3		
	DBStd	6.78	2.92	2.71	3.63	3.96	4.22	4.21	4.27	4.88	5.91	5.01	4.95	3.61		
	HDD50	28	0	0	0	0	3	8	14	2	1	0	0	0		
	HDD65	1338	2	2	20	74	189	298	347	199	93	62	41	11		
	CDD50	4869	612	545	534	401	280	160	132	272	406	467	491	569		
	CDD65	703	149	127	89	25	2	0	0	4	48	63	82	114		
	CDH74	3652	602	511	306	119	44	11	8	132	533	476	397	513		
	CDH80	558	87	73	30	8	1	0	0	7	113	93	60	86		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	32.6	5.6	4.9	4.1	1.8	0.6	0.3	0.3	0.4	0.9	3.1	5.0	5.8		
	PrecMax	50.6	10.7	13.8	7.3	3.6	1.6	1.1	1.0	1.1	3.2	5.4	10.0	10.3		
	PrecMin	17.2	1.7	1.5	0.9	0.2	0.0	0.0	0.0	0.0	0.1	1.5	1.9	2.4		
	PrecStd	7.5	2.4	3.3	1.7	0.8	0.5	0.3	0.3	0.3	0.7	1.1	1.7	1.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.4	86.1	83.6	80.8	77.6	75.7	74.7	81.0	86.7	86.3	84.9	86.2		
		MCWB	64.8	64.5	63.5	59.9	56.2	53.1	50.3	56.4	57.1	59.6	62.3	64.8		
	2%	DB	82.3	82.5	80.3	77.2	74.8	71.7	70.8	77.9	83.1	82.7	81.6	82.4		
		MCWB	66.8	65.4	63.8	60.2	55.3	51.3	49.3	53.8	56.3	59.8	62.5	64.3		
	5%	DB	79.8	79.9	77.8	74.6	72.1	68.3	67.1	74.8	80.1	79.3	78.6	79.6		
		MCWB	65.7	64.5	63.6	59.8	54.5	50.4	48.4	53.0	56.0	59.6	62.3	65.1		
	10%	DB	77.4	77.4	75.1	71.7	68.9	65.0	63.5	70.8	76.5	75.9	75.5	76.8		
		MCWB	65.5	64.0	62.9	59.0	53.4	49.4	47.6	51.6	55.2	59.3	62.4	64.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.0	69.5	69.7	64.8	59.5	54.9	53.3	59.3	61.9	66.9	69.0	69.2		
		MCDB	80.7	79.8	74.9	74.6	72.8	68.6	65.3	74.0	79.6	73.8	74.2	79.3		
	2%	WB	68.3	67.5	67.2	63.2	57.6	53.3	51.3	56.7	60.0	64.6	66.7	67.8		
		MCDB	78.4	77.5	73.7	71.6	69.8	67.3	65.2	70.1	76.7	73.6	73.2	76.8		
	5%	WB	67.4	66.3	65.8	61.9	56.2	51.9	49.9	55.4	58.5	62.9	65.2	66.4		
		MCDB	76.8	76.1	73.0	70.0	67.3	64.5	63.2	68.4	74.0	72.0	73.0	74.8		
	10%	WB	66.4	65.2	64.5	60.6	54.8	50.6	48.6	53.8	56.7	61.3	64.1	65.4		
		MCDB	75.2	74.3	71.8	68.5	65.1	61.9	60.3	66.4	71.4	70.9	72.2	73.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.8	17.2	16.5	19.0	23.4	24.0	24.1	25.1	25.2	20.4	16.5	16.1		
		MCDBR	20.7	21.7	21.5	22.6	26.2	27.3	27.6	29.3	29.8	26.5	22.5	21.8		
	5% WB	MCWBR	8.3	7.6	8.0	8.4	10.4	11.3	11.4	11.4	10.7	9.1	8.0	8.1		
		MCWBR	17.1	18.6	17.2	18.5	22.9	23.9	24.7	25.1	26.0	20.3	17.1	17.1		
(40) Clear-Sky Solar Irradiance	taub		0.375	0.369	0.363	0.339	0.305	0.288	0.289	0.344	0.393	0.418	0.384	0.387		
	taud		2.401	2.417	2.421	2.451	2.508	2.534	2.516	2.351	2.244	2.220	2.353	2.361		
	Ebn at Noon		307	304	297	290	288	287	291	283	282	286	302	304		
	Edh at Noon		41	39	37	33	28	26	28	36	43	47	42	42		
(44) All-Sky Solar Radiation	RadAvg		1962	1918	1708	1464	1370	1243	1330	1542	1756	1797	1884	1938		
	RadStd		155	176	150	113	66	75	41	82	106	124	168	131		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page